

LAB ACTIVITY (Lab-05)

/* Multiplication Table Generator

Write a program to input a number n and print its multiplication table up to 10 using all four loops (for, while, do-while, for-each). */

```
import java.util.Scanner;

public class multiplication {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int n = sc.nextInt();

        System.out.println("\n--- Using for loop ---");
        for (int i = 1; i <= 10; i++) {
            System.out.println(n + " x " + i + " = " + (n * i));
        }

        System.out.println("\n--- Using while loop ---");
        int i = 1;
        while (i <= 10) {
            System.out.println(n + " x " + i + " = " + (n * i));
            i++;
        }

        System.out.println("\n--- Using do-while loop ---");
        i = 1;
        do {
            System.out.println(n + " x " + i + " = " + (n * i));
            i++;
        } while (i <= 10);

        System.out.println("\n--- Using for-each loop ---");
```

```
int[] nums = {1,2,3,4,5,6,7,8,9,10};
for (int num : nums) {
    System.out.println(n + " x " + num + " = " + (n * num));
}

sc.close();
}
}
```

```
D:\SEMESTER I\LAB WORK\Java Programming\lab5(0609)>javac multiplication.java
```

```
D:\SEMESTER I\LAB WORK\Java Programming\lab5(0609)>java multiplication
```

```
Enter a number: 3
```

```
--- Using for loop ---
```

```
3 x 1 = 3
3 x 2 = 6
3 x 3 = 9
3 x 4 = 12
3 x 5 = 15
3 x 6 = 18
3 x 7 = 21
3 x 8 = 24
3 x 9 = 27
3 x 10 = 30
```

```
--- Using while loop ---
```

```
3 x 1 = 3
3 x 2 = 6
3 x 3 = 9
3 x 4 = 12
3 x 5 = 15
3 x 6 = 18
3 x 7 = 21
3 x 8 = 24
3 x 9 = 27
3 x 10 = 30
```

```
--- Using do-while loop ---
```

```
3 x 1 = 3
3 x 2 = 6
3 x 3 = 9
3 x 4 = 12
3 x 5 = 15
3 x 6 = 18
3 x 7 = 21
3 x 8 = 24
3 x 9 = 27
3 x 10 = 30
```

```
--- Using for-each loop ---
```

```
3 x 1 = 3
3 x 2 = 6
3 x 3 = 9
3 x 4 = 12
3 x 5 = 15
3 x 6 = 18
3 x 7 = 21
3 x 8 = 24
3 x 9 = 27
3 x 10 = 30
```

// Factorial of a Number

Write a program to compute the factorial of a given number using all four loops

```
import java.util.Scanner;

public class FactorialUsingLoops {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int n = sc.nextInt();

        // --- Using for loop ---
        long factFor = 1;
        for (int i = 1; i <= n; i++) {
            factFor *= i;
        }
        System.out.println("Factorial using for loop: " + factFor);

        // --- Using while loop ---
        long factWhile = 1;
        int i = 1;
        while (i <= n) {
            factWhile *= i;
            i++;
        }
        System.out.println("Factorial using while loop: " + factWhile);

        // --- Using do-while loop ---
        long factDoWhile = 1;
        i = 1;
        if (n == 0) { // special case for 0!
```

```

        factDoWhile = 1;
    } else {
        do {
            factDoWhile *= i;
            i++;
        } while (i <= n);
    }
    System.out.println("Factorial using do-while loop: " + factDoWhile);

    // --- Using for-each loop ---
    long factForEach = 1;
    int[] numbers = new int[n];
    for (int j = 0; j < n; j++) {
        numbers[j] = j + 1;
    }
    for (int num : numbers) {
        factForEach *= num;
    }
    System.out.println("Factorial using for-each loop: " + factForEach);

    sc.close();
}
}

```

```

D:\SEMESTER I\LAB WORK\Java Programming\lab5(0609)>javac FactorialUsingLoops.java
D:\SEMESTER I\LAB WORK\Java Programming\lab5(0609)>java FactorialUsingLoops
Enter a number: 5
Factorial using for loop: 120
Factorial using while loop: 120
Factorial using do-while loop: 120
Factorial using for-each loop: 120

```

// Reverse a Number

Write a program to reverse the digits of a number using for, while, do-while, and for-each.

```
import java.util.Scanner;

public class reverse {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int n = sc.nextInt();

        // --- Using while loop ---
        int num = n;
        int revWhile = 0;
        while (num > 0) {
            int digit = num % 10;
            revWhile = revWhile * 10 + digit;
            num /= 10;
        }
        System.out.println("Reverse using while loop: " + revWhile);

        // --- Using do-while loop ---
        num = n;
        int revDoWhile = 0;
        if (num == 0) {
            revDoWhile = 0;
        } else {
            do {
                int digit = num % 10;
                revDoWhile = revDoWhile * 10 + digit;
                num /= 10;
            } while (num > 0);
        }
        System.out.println("Reverse using do-while loop: " + revDoWhile);

        // --- Using for loop ---
        num = n;
        int revFor = 0;
```

```
for (; num > 0; num /= 10) {
    int digit = num % 10;
    revFor = revFor * 10 + digit;
}
System.out.println("Reverse using for loop: " + revFor);

// --- Using for-each loop ---
String strNum = Integer.toString(n);
char[] digits = strNum.toCharArray();
String revForEach = "";
for (char d : digits) {
    revForEach = d + revForEach; // prepend digit
}
System.out.println("Reverse using for-each loop: " + revForEach);

sc.close();
}
}
```

```
D:\SEMESTER I\LAB WORK\Java Programming\lab5(0609)>javac reverse.java
D:\SEMESTER I\LAB WORK\Java Programming\lab5(0609)>java reverse
Enter a number: 15678
Reverse using while loop: 87651
Reverse using do-while loop: 87651
Reverse using for loop: 87651
Reverse using for-each loop: 87651
```

Fibonacci Series

Write a program to generate the first n terms of the Fibonacci sequence using all four loop types

```
import java.util.Scanner;

public class fibonacci {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter number of terms (n): ");
        int n = sc.nextInt();

        // --- Using for loop ---
        System.out.println("\nFibonacci using for loop:");
        int a = 0, b = 1;
        for (int i = 1; i <= n; i++) {
            System.out.print(a + " ");
            int next = a + b;
            a = b;
            b = next;
        }

        // --- Using while loop ---
        System.out.println("\n\nFibonacci using while loop:");
        a = 0; b = 1;
        int i = 1;
        while (i <= n) {
            System.out.print(a + " ");
            int next = a + b;
            a = b;
            b = next;
            i++;
        }

        // --- Using do-while loop ---
        System.out.println("\n\nFibonacci using do-while loop:");
        a = 0; b = 1;
        i = 1;
        if (n > 0) {
```

```

        do {
            System.out.print(a + " ");
            int next = a + b;
            a = b;
            b = next;
            i++;
        } while (i <= n);
    }

    // --- Using for-each loop ---
    System.out.println("\n\nFibonacci using for-each loop:");
    int[] fibArray = new int[n];
    if (n > 0) fibArray[0] = 0;
    if (n > 1) fibArray[1] = 1;
    for (int j = 2; j < n; j++) {
        fibArray[j] = fibArray[j - 1] + fibArray[j - 2];
    }
    for (int term : fibArray) {
        System.out.print(term + " ");
    }

    sc.close();
}
}

```

```

D:\SEMESTER I\LAB WORK\Java Programming\lab5(0609)>java fibonacci
Enter number of terms (n): 6

Fibonacci using for loop:
0 1 1 2 3 5

Fibonacci using while loop:
0 1 1 2 3 5

Fibonacci using do-while loop:
0 1 1 2 3 5

Fibonacci using for-each loop:
0 1 1 2 3 5

```

Sum of Digits

Write a program to calculate the sum of digits of a number using all loop types

```
import java.util.Scanner;

public class SumOfDigits {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int n = sc.nextInt();

        // --- Using while loop ---
        int num = n;
        int sumWhile = 0;
        while (num > 0) {
            int digit = num % 10;
            sumWhile += digit;
            num /= 10;
        }
        System.out.println("Sum using while loop: " + sumWhile);

        // --- Using do-while loop ---
        num = n;
        int sumDoWhile = 0;
        if (num == 0) {
            sumDoWhile = 0;
        } else {
            do {
                int digit = num % 10;
                sumDoWhile += digit;
                num /= 10;
            } while (num > 0);
        }
        System.out.println("Sum using do-while loop: " + sumDoWhile);
    }
}
```

```
// --- Using for loop ---
    num = n;
    int sumFor = 0;
    for (; num > 0; num /= 10) {
        int digit = num % 10;
        sumFor += digit;
    }
    System.out.println("Sum using for loop: " + sumFor);

    // --- Using for-each loop ---
    String strNum = Integer.toString(n);
    char[] digits = strNum.toCharArray();
    int sumForEach = 0;
    for (char d : digits) {
        sumForEach += Character.getNumericValue(d);
    }
    System.out.println("Sum using for-each loop: " + sumForEach);

    sc.close();
}
}
```

```
D:\SEMESTER I\LAB WORK\Java Programming\lab5(0609)>java SumOfDigits
Enter a number: 3456
Sum using while loop: 18
Sum using do-while loop: 18
Sum using for loop: 18
Sum using for-each loop: 18
```

Find Maximum and Minimum in an Array

Write a program to input an array and determine the maximum and minimum element using for, while, do-while, and for-each

```
import java.util.Scanner;

public class MaxMinArray {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        // Input array
        System.out.print("Enter size of array: ");
        int n = sc.nextInt();
        int arr[] = new int[n];

        System.out.println("Enter " + n + " elements:");
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        // ----- Using FOR loop -----
        int maxFor = arr[0], minFor = arr[0];
        for (int i = 1; i < n; i++) {
            if (arr[i] > maxFor) maxFor = arr[i];
            if (arr[i] < minFor) minFor = arr[i];
        }
        System.out.println("FOR loop -> Max: " + maxFor + ", Min: " + minFor);

        // ----- Using WHILE loop -----
        int maxWhile = arr[0], minWhile = arr[0];
        int i = 1;
        while (i < n) {
            if (arr[i] > maxWhile) maxWhile = arr[i];
            if (arr[i] < minWhile) minWhile = arr[i];
            i++;
        }
    }
}
```

```
System.out.println("WHILE loop -> Max: " + maxWhile + ", Min: " + minWhile);
```

```
// ----- Using DO-WHILE loop -----
int maxDo = arr[0], minDo = arr[0];
int j = 1;
do {
    if (arr[j] > maxDo) maxDo = arr[j];
    if (arr[j] < minDo) minDo = arr[j];
    j++;
} while (j < n);
System.out.println("DO-WHILE loop -> Max: " + maxDo + ", Min: " + minDo);
```

```
// ----- Using FOR-EACH loop -----
int maxFE = arr[0], minFE = arr[0];
for (int val : arr) {
    if (val > maxFE) maxFE = val;
    if (val < minFE) minFE = val;
}
System.out.println("FOR-EACH loop -> Max: " + maxFE + ", Min: " + minFE);
```

```
sc.close();
}
}
```

```
D:\SEMESTER I\LAB WORK\Java Programming\lab5(0609)>javac MaxMinArray.java
D:\SEMESTER I\LAB WORK\Java Programming\lab5(0609)>java MaxMinArray
Enter size of array: 5
Enter 5 elements:
3
5
7
1
9
FOR loop -> Max: 9, Min: 1
WHILE loop -> Max: 9, Min: 1
DO-WHILE loop -> Max: 9, Min: 1
FOR-EACH loop -> Max: 9, Min: 1
```

Prime Number Check

Write a program to check whether a given number is prime or not using all loop variations.

```
import java.util.Scanner;

public class PrimeCheck {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        // Input number
        System.out.print("Enter a number: ");
        int n = sc.nextInt();

        // ----- Using FOR loop -----
        boolean isPrimeFor = true;
        if (n <= 1) {
            isPrimeFor = false;
        } else {
            for (int i = 2; i <= n / 2; i++) {
                if (n % i == 0) {
                    isPrimeFor = false;
                    break;
                }
            }
        }
        System.out.println("FOR loop -> " + (isPrimeFor ? "Prime" : "Not Prime"));

        // ----- Using WHILE loop -----
        boolean isPrimeWhile = true;
        if (n <= 1) {
            isPrimeWhile = false;
        } else {
            int i = 2;
            while (i <= n / 2) {
                if (n % i == 0) {
                    isPrimeWhile = false;
                }
            }
        }
    }
}
```

```

        break;
    }
    i++;
}
}
System.out.println("WHILE loop -> " + (isPrimeWhile ? "Prime" : "Not
Prime"));

```

```

// ----- Using DO-WHILE loop -----
boolean isPrimeDo = true;
if (n <= 1) {
    isPrimeDo = false;
} else {
    int i = 2;
    do {
        if (n % i == 0) {
            isPrimeDo = false;
            break;
        }
        i++;
    } while (i <= n / 2);
}
System.out.println("DO-WHILE loop -> " + (isPrimeDo ? "Prime" : "Not
Prime"));

```

```

// ----- Using FOR-EACH loop -----

```

```

boolean isPrimeFE = true;
if (n <= 1) {
    isPrimeFE = false;
} else {
    int[] divisors = new int[n / 2 - 1];
    for (int i = 0; i < divisors.length; i++) {
        divisors[i] = i + 2;
    }

    for (int d : divisors) {
        if (n % d == 0) {
            isPrimeFE = false;
            break;
        }
    }
}

```

```

    }
}
System.out.println("FOR-EACH loop -> " + (isPrimeFE ? "Prime" : "Not
Prime"));

    sc.close();
}
}

```

```

D:\SEMESTER I\LAB WORK\Java Programming\lab5(0609)>javac PrimeCheck.java

D:\SEMESTER I\LAB WORK\Java Programming\lab5(0609)>java PrimeCheck
Enter a number: 79
FOR loop -> Prime
WHILE loop -> Prime
DO-WHILE loop -> Prime
FOR-EACH loop -> Prime

```

Palindrome Checker

Write a program to check whether a string or number is a palindrome using different loops.

```

import java.util.Scanner;

public class PalindromeChecker {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        // Input string/number

        System.out.print("Enter a string or number: ");

        String input = sc.nextLine();

        // Normalize input (optional, in case user types spaces/case differences)

        String str = input.toLowerCase().replaceAll("\\s+", "");
    }
}

```

```
// ----- Using FOR loop -----

boolean isPalindromeFor = true;

for (int i = 0; i < str.length() / 2; i++) {

    if (str.charAt(i) != str.charAt(str.length() - 1 - i)) {

        isPalindromeFor = false;

        break;

    }

}

System.out.println("FOR loop -> " + (isPalindromeFor ? "Palindrome" : "Not Palindrome"));


// ----- Using WHILE loop -----

boolean isPalindromeWhile = true;

int left = 0, right = str.length() - 1;

while (left < right) {

    if (str.charAt(left) != str.charAt(right)) {

        isPalindromeWhile = false;

        break;

    }

    left++;

    right--;

}

System.out.println("WHILE loop -> " + (isPalindromeWhile ? "Palindrome" : "Not Palindrome"));


// ----- Using DO-WHILE loop -----

boolean isPalindromeDo = true;

int l = 0, r = str.length() - 1;

if (str.length() > 1) { // avoid empty string issue

    do {

        if (str.charAt(l) != str.charAt(r)) {

            isPalindromeDo = false;

        }

    } while (l < r);

}
```

```

        break;

    }

    l++;

    r--;

} while (l < r);

}

System.out.println("DO-WHILE loop -> " + (isPalindromeDo ? "Palindrome" : "Not
Palindrome"));

// ----- Using FOR-EACH loop -----

StringBuilder reversed = new StringBuilder();

for (char c : str.toCharArray()) {

    reversed.insert(0, c); // build reversed string

}

boolean isPalindromeFE = str.equals(reversed.toString());

System.out.println("FOR-EACH loop -> " + (isPalindromeFE ? "Palindrome" : "Not
Palindrome"));

sc.close();

}

}

```

```

D:\SEMESTER I\LAB WORK\Java Programming\lab5(0609)>javac PalindromeChecker.java

D:\SEMESTER I\LAB WORK\Java Programming\lab5(0609)>java PalindromeChecker
Enter a string or number: 121madam121
FOR loop -> Palindrome
WHILE loop -> Palindrome
DO-WHILE loop -> Palindrome
FOR-EACH loop -> Palindrome

```

Matrix Traversal

Write a program to input a 2D matrix and print its elements row-wise and column-wise using for, while, do-while, and for-each.

```
import java.util.Scanner;

public class MatrixTraversal {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        // Input matrix size

        System.out.print("Enter number of rows: ");

        int rows = sc.nextInt();

        System.out.print("Enter number of columns: ");

        int cols = sc.nextInt();

        int[][] matrix = new int[rows][cols];

        // Input matrix elements

        System.out.println("Enter elements of matrix:");

        for (int i = 0; i < rows; i++) {

            for (int j = 0; j < cols; j++) {

                matrix[i][j] = sc.nextInt();

            }

        }

        System.out.println("\n--- Row-wise Traversal ---");

        // ----- FOR loop -----

        System.out.println("FOR loop:");

        for (int i = 0; i < rows; i++) {
```

```
        for (int j = 0; j < cols; j++) {  
            System.out.print(matrix[i][j] + " ");  
        }  
        System.out.println();  
    }  
  
    // ----- WHILE loop -----  
    System.out.println("WHILE loop:");  
    int i = 0;  
    while (i < rows) {  
        int j = 0;  
        while (j < cols) {  
            System.out.print(matrix[i][j] + " ");  
            j++;  
        }  
        System.out.println();  
        i++;  
    }  
  
    // ----- DO-WHILE loop -----  
    System.out.println("DO-WHILE loop:");  
    int r = 0;  
    if (rows > 0 && cols > 0) {  
        do {  
            int c = 0;  
            do {  
                System.out.print(matrix[r][c] + " ");  
                c++;  
            } while (c < cols);  
            System.out.println();  
            r++;  
        } while (r < rows);  
    }
```

```
        } while (r < rows);
    }

    // ----- FOR-EACH loop -----
    System.out.println("FOR-EACH loop:");
    for (int[] row : matrix) {
        for (int val : row) {
            System.out.print(val + " ");
        }
        System.out.println();
    }

    System.out.println("\n--- Column-wise Traversal ---");

    // ----- FOR loop -----
    System.out.println("FOR loop:");
    for (int j = 0; j < cols; j++) {
        for (int k = 0; k < rows; k++) {
            System.out.print(matrix[k][j] + " ");
        }
        System.out.println();
    }

    // ----- WHILE loop -----
    System.out.println("WHILE loop:");
    int c1 = 0;
    while (c1 < cols) {
        int r1 = 0;
        while (r1 < rows) {
            System.out.print(matrix[r1][c1] + " ");
            r1++;
        }
    }
}
```

```
    }

    System.out.println();

    c1++;
}

// ----- DO-WHILE loop -----

System.out.println("DO-WHILE loop:");

int c2 = 0;

if (rows > 0 && cols > 0) {

    do {

        int r2 = 0;

        do {

            System.out.print(matrix[r2][c2] + " ");

            r2++;

        } while (r2 < rows);

        System.out.println();

        c2++;

    } while (c2 < cols);

}

// ----- FOR-EACH loop (Trick: Not direct, so transpose logic) -----

System.out.println("FOR-EACH loop:");

for (int j = 0; j < cols; j++) {

    for (int[] row : matrix) {

        System.out.print(row[j] + " ");

    }

    System.out.println();

}

sc.close();

}
```

```
D:\SEMESTER I\LAB WORK\Java Programming\lab5(0609)>javac MatrixTraversal.java

D:\SEMESTER I\LAB WORK\Java Programming\lab5(0609)>java MatrixTraversal
Enter number of rows: 2
Enter number of columns: 3
Enter elements of matrix:
1 2 3
4 5 6

--- Row-wise Traversal ---
FOR loop:
1 2 3
4 5 6
WHILE loop:
1 2 3
4 5 6
DO-WHILE loop:
1 2 3
4 5 6
FOR-EACH loop:
1 2 3
4 5 6

--- Column-wise Traversal ---
FOR loop:
1 4
2 5
3 6
WHILE loop:
1 4
2 5
3 6
DO-WHILE loop:
1 4
2 5
3 6
FOR-EACH loop:
1 4
2 5
3 6
```

Pattern Printing

Write a program to print the following pattern using all loops:

*

**

```
import java.util.Scanner;

public class PatternPrinting {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        // Input number of rows

        System.out.print("Enter number of rows: ");

        int n = sc.nextInt();

        // ----- FOR loop -----

        System.out.println("FOR loop:");

        for (int i = 1; i <= n; i++) {

            for (int j = 1; j <= i; j++) {

                System.out.print("*");

            }

            System.out.println();

        }

    }

}
```

```
// ----- WHILE loop -----
```

```
System.out.println("WHILE loop:");
```

```
int i = 1;
```

```
while (i <= n) {
```

```
    int j = 1;
```

```
    while (j <= i) {
```

```
        System.out.print("*");
```

```
        j++;
```

```
    }
```

```
    System.out.println();
```

```
    i++;
```

```
}
```

```
// ----- DO-WHILE loop -----
```

```
System.out.println("DO-WHILE loop:");
```

```
int r = 1;
```

```
do {
```

```
    int c = 1;
```

```
    do {
```

```
        System.out.print("*");
```

```
        c++;
```

```
    } while (c <= r);
```

```
    System.out.println();
```

```
    r++;
```

```
} while (r <= n);
```

```
// ----- FOR-EACH loop (Trick: simulate with array of row numbers) -----
```

```
System.out.println("FOR-EACH loop:");
```

```
int[] rows = new int[n];
```

```

    for (int k = 0; k < n; k++) {
        rows[k] = k + 1;
    }
    for (int row : rows) {
        for (int j = 1; j <= row; j++) {
            System.out.print("*");
        }
        System.out.println();
    }

    sc.close();
}
}

```

```
D:\SEMESTER I\LAB WORK\Java Programming\lab5(0609)>javac PatternPrinting.java
```

```
D:\SEMESTER I\LAB WORK\Java Programming\lab5(0609)>java PatternPrinting
```

```
Enter number of rows: 4
```

```
FOR loop:
```

```
*
**
***
****
```

```
WHILE loop:
```

```
*
**
***
****
```

```
DO-WHILE loop:
```

```
*
**
***
****
```

```
FOR-EACH loop:
```

```
*
**
***
****
```